A prospective study of intake of trans-fatty acids from ruminant fat, partially hydrogenated vegetable oils, and marine oils and mortality from CVD.

Trans-fatty acids (TFA) have adverse effects on blood lipids, but whether TFA from different sources are associated with risk of CVD remains unresolved. The objective of the present study was to evaluate the association between TFA intake from partially hydrogenated vegetable oils (PHVO), partially hydrogenated fish oils (PHFO) and ruminant fat (rTFA) and risks of death of CVD, CHD, cerebrovascular diseases and sudden death in the Norwegian Counties Study, a population-based cohort study. Between 1974 and 1988, participants were examined for up to three times. Fat intake was assessed with a semi-quantitative FFQ. A total of 71,464 men and women were followed up through 2007. Hazard ratios (HR) and 95 % CI were estimated with Cox regression. Energy from TFA was compared to energy from all other sources, carbohydrates or unsaturated cis-fatty acids with different multivariable models. During follow-up, 3870 subjects died of CVD, 2383 of CHD, 732 of cerebrovascular diseases and 243 of sudden death. Significant risks, comparing highest to lowest intake category, were found for: TFA from PHVO and CHD (HR 1.23 (95 % CI 1.00, 1.50)) and cerebrovascular diseases (HR 0.65 (95 % CI 0.45, 0.94)); TFA from PHFO and CVD (HR 1.14 (95 % CI 1.03, 1.26)) and cerebrovascular diseases (HR 1.32 (95 % CI 1.04, 1.69)); and rTFA intake and CVD (HR 1.30 (95 % CI 1.05, 1.61)), CHD (HR 1.50 (95 % CI 1.11, 2.03)) and sudden death (HR 2.73 (95 % CI 1.19, 6.25)) in women. These associations with rTFA intake were not significant in men (P interaction ≥ 0.01). The present study supports that TFA intake, irrespective of source, increases CVD risk. Whether TFA from PHVO decreases risk of cerebrovascular diseases warrants further investigation.